MIL-E-52171
27 April 1961
(See sec. 6)

MILITARY SPECIFICATION

ETHYLENE OXIDE, TECHNICAL

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers one grade of ethylene oxide.

2. APPLICABLE DOCUMENTS

2.1 **Government documents.** The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

O-E-670—Ethyl Alcohol (Ethanol); Denatured Alcohol, and Proprietary Solvent.

O-M-232—Methanol, Technical, (Methyl Alcohol).

STANDARDS

MILITARY

MIL-STD-129—Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING MATERIALS

ASTM Standard

D1364—Method for Testing for Water in Lacquer Solvents and Diluents.

(Applications for copies of the ASTM publications should be addressed to the American Society for Testing Materials, 1916 Race St., Philadelphia, Pa.)

INTERSTATE COMMERCE COMMISSION

49 CFR 71-90—Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission Regulations are a part of the Code of Federal Regulations (1949 Edition Revised 1956) available from the Superintendent of Documents, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-90 (Rev. 1956).")

3. REQUIREMENTS

3.1 **Chemical composition.** The chemical composition of the ethylene oxide shall con-

TABLE I. *Chemical Composition*

Composition	Percent by weight		Test paragraph
	Minimum	Maximum	
Assay [$(CH_2)_2O$]	99.0	.....	4.5.1
Water	.....	0.030	4.5.2
Acidity (as acetic acid)	.....	0.002	4.5.3
Aldehydes (as acetaldehyde)	.....	0.010	4.5.4
Chlorides (inorganic)	.....	none	4.5.5
Acetylene	.....	none	4.5.6

FSC 6830

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

form to the percent by weight requirements specified in table I when tested in accordance with the test paragraphs referenced therein.

3.2 **Nonvolatile residue.** The ethylene oxide shall contain not more than 0.010 grams (g) of residue per 100 milliliters (ml) when tested as specified in 4.5.7.

3.3 **Color.** The color of the ethylene oxide shall be not greater than 15 on the platinum-cobalt scale when tested as specified in 4.5.8.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein, the supplier shall be responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 **Special provisions.**

4.2.1 *Alternative inspection (including testing) procedures.* The supplier may utilize any alternative inspection procedures which will insure equal or better quality by submitting a written proposal with justification and obtaining written approval from the Government prior to instituting the procedure. In case of dispute, the procedures of this specification will govern.

4.2.2 *Supplemental evidence.* The supplier shall provide evidence acceptable to the contracting officer that the requirements of section 5 have been satisfied.

4.3 **Lotting.** A lot shall consist of the ethylene oxide offered for acceptance at one time, which has been produced by one manufacturer under essentially the same manufacturing conditions, and with no change in materials provided the process is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3).

4.4 **Sampling.** Sample containers shall be taken at random from each lot as follows:

Lot size	*Sample size*
1	1
2–275	2
276–545	3
546–900	4
901–1345	5
1346–1875	6
1876–2500	7

Sample size for lots exceeding 2500 containers shall be calculated using the following equation:

$$n = 0.15\sqrt{N}$$

where: n = sample size.
N = lot size.

Using recommended safety precautions and procedures, a 1-liter specimen shall be taken from each sample container and place in a chilled, clean, dry container labeled to identify the specimen with the lot and the sample container from which it was taken. The specimens shall immediately be placed in an ice bath and maintained at a temperature of approximately 1° C. A 300-ml portion of each specimen shall be blended into a composite specimen and placed in a chilled, clean, dry container. The composite and the individual specimens shall be maintained at approximately 1° C. until the tests are completed. The composite and one of the individual specimens, selected at random, shall be tested separately as specified in 4.5.

4.5 **Tests.** Tests shall be conducted as follows: Exercise care in the handling of ethylene oxide as it is an explosive and toxic material (see 6.4 and 6.5). Conduct all tests in a ventilated hood. Use analytical grade reagents and distilled water throughout the tests. Run blank determinations in parallel with the tests, using the same quantities of reagents, and apply corrections when significant. Determine the specific gravity of the specimen at 1° ± 1° C. by means of a suitable hydrometer, and use the figure obtained for calculations.

4.5.1 *Assay* [$(CH_2)_2O$].

4.5.1.1 *Preparation of specimen.* Suspend a sample bottle and metal carrier similar to figure 1 in a dry ice-acetone bath contained in a Dewar flask. Fill the bottle to the shoulder with a portion of the sample ethylene oxide and cover the bottle with a glass

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:09Z
Check the source to verify that this is the current version before use.

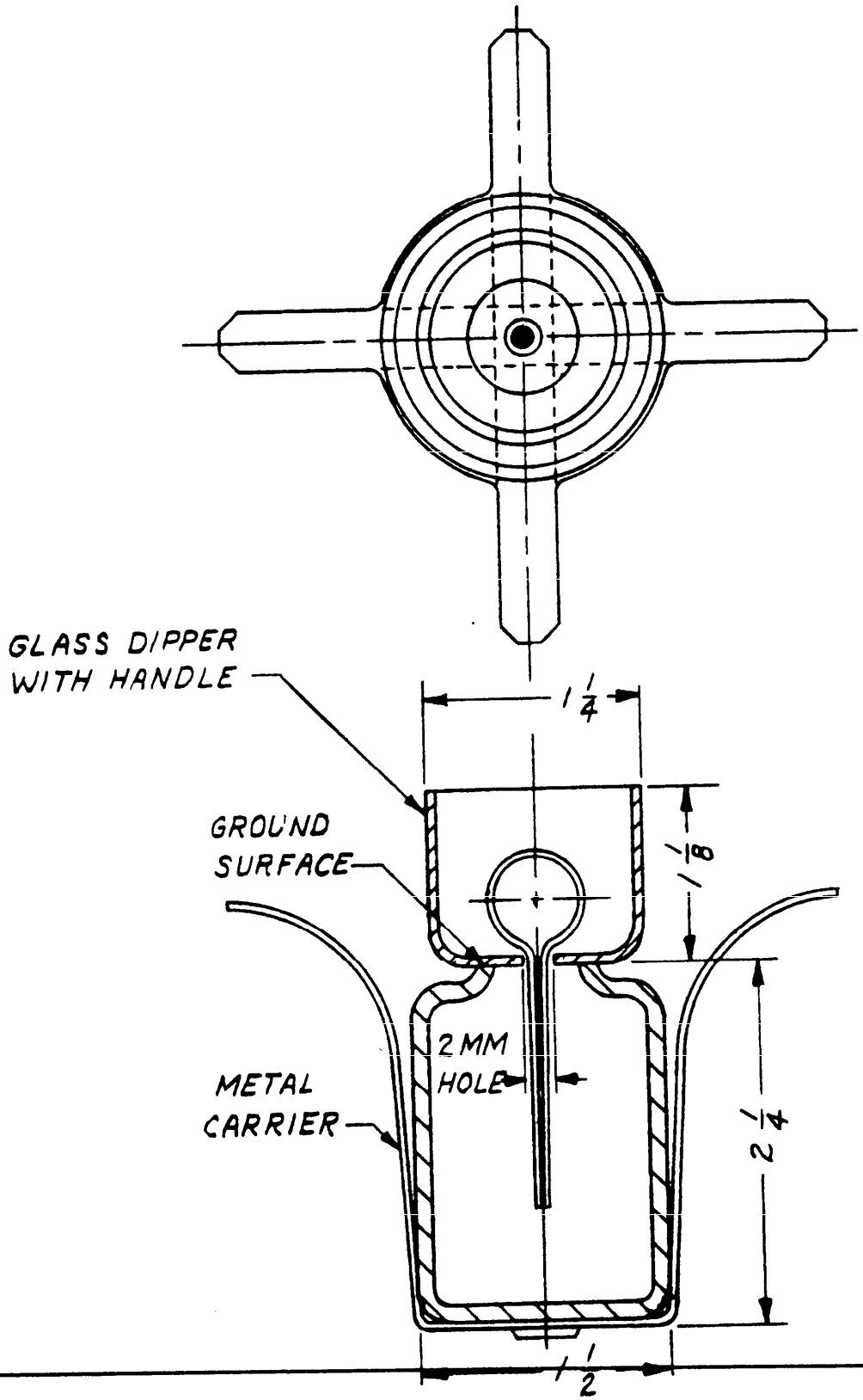


FIGURE 1. *Bottle and metal carrier.*

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:09Z
Check the source to verify that this is the current version before use.

plate. Dry a borosilicate glass bulb 18 to 22 millimeters in diameter having a narrow stem approximately 50 millimeters long in an oven at 105° ± 5° C, cool it in a desiccator, and weigh it to the nearest 0.1 milligram (mg). Place the bulb in the glass dipper, and replace the glass plate with the bulb and dipper as shown on figure 1. Cover the bulb in the glass dipper with carbon dioxide (CO_2) snow, and allow sufficient time to elapse for 1 g of the sample to be drawn into the bulb. Immediately remove the glass dipper by the handle, invert the bulb, and then hold the bulb stem at an angle of approximately 45 degrees from the vertical. Bring the stem near a gas flame to warm slightly. When the liquid leaves the capillary, seal the bulb quickly with the gas flame while the bulb is still in the CO_2 snow.

4.5.1.2 *Procedure.* Remove the bulb from the glass dipper. Allow the bulb to come to room temperature in a desiccator and weigh to the nearest 0.1 mg. Place 150 g of magnesium chloride, 125 ml of saturated magnesium chloride solution, and 30.0 ml of 1*N* sulfuric acid in a 500-ml heavy-wall, glass-stoppered erlenmeyer flask. (Thirty milliequivalents of sulfuric acid are approximately equal to 1.34 g of 99.0 percent ethylene oxide. If the normality of the sulfuric acid is much less than 1*N*, an additional quantity of acid should be added to make a total of 30 milliequivalents. Also, if the sample is 1.34 g or more, additional acid and magnesium chloride should be added.) Introduce the sealed sample of ethylene oxide into the erlenmeyer flask and stopper tightly, taking care not to break the bulb. (Since the flask must be sealed, the stopper should be reground with No. 300 carborundum and greased.) Cool in an ice bath to 1° ± 1° C. Remove the flask, wrap with a towel, and holding the stopper down tightly, shake to break the bulb. Allow the contents to react, shaking intermittently for 15 minutes. If the bulb stem is not completely shattered by shaking, crush the stem with a stirring rod. Titrate the excess acid with 0.1*N* sodium hydroxide to a bromcresol purple endpoint (yellow to blue). Run a blank titration on the reagents, using exactly the same amount of sulfuric acid.

4.5.1.3 *Calculation.*

$$\text{Percent ethylene oxide} = \frac{(B - A)\ N \times 4.405}{\text{g of sample}}$$

where: A = ml of sodium hydroxide solution required for the sample.
B = ml sodium hydroxide solution required for the blank.
C = normality of the sodium hydroxide solution.

4.5.1.4 *Retest.* Retest the lot if the results of tests of the composite and the individual samples vary over a range greater than 0.3 percent.

4.5.2 *Water.* Determine the water content of a 50-ml specimen of the ethylene oxide as specified in ASTM Standard D1364. Calculate the weight of the specimen tested using the specific gravity of the ethylene oxide determined in 4.5.

4.5.3 *Acidity (as acetic acid).* Using a chilled 100 ml graduate, measure a 68 ml (60 g) portion of the sample into a 250 ml erlenmeyer flask. Allow the portion to evaporate to about 50 ml to release any carbon dioxide in the sample. Immediately add 3 drops of a 1.0 percent solution of phenolphthalein in a 1:1 mixture of methanol and water, and titrate with an aqueous 0.01*N* sodium hydroxide to a pink end point which persists for 15 seconds. Calculate the acidity as acetic acid using the formula:

$$\text{Percent acidity (as acetic acid)} = \frac{A \times N \times 6.0}{B \times G}$$

where: A = ml of 0.01*N* sodium hydroxide
N = normality of sodium hydroxide
B = ml of ethylene oxide
G = specific gravity of sample

4.5.4 *Aldehydes (as acetaldehyde).*

4.5.4.1 *Reagents.*

4.5.4.1.1 *Sodium metabisulfite solution (0.1N).* Dissolve 1.75 to 5.00 g of sodium metabisulfite in sufficient water to make 1,000 ml of solution. Adjust the pH of the solution

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

to 3.5 by adding 0.5*N* sulfuric acid. Make up a fresh solution daily. Do not use this solution after the pH drops below 3.0.

4.5.4.1.2 *Iodine solution* (0.1*N*). Dissolve 13.5 g of iodine in 200 ml of water containing 24 g of potassium iodide. Dilute the solution to 1,000 ml with water. Standardize the solution against standard sodium thiosulfate or standard sodium arsenite solutions.

4.5.4.2 *Procedure.* Transfer 25.0 ml of distilled water to each of four 500-ml glass-stoppered erlenmeyer flasks and cool them to approximately 0° C. Carefully pipet 20.0 ml of the 0.1*N* sodium metabisulfate solution into each flask, using the same pipet for each transfer and holding the tip of the pipet under the surface of the water in the flask during delivery. Cool the contents of the flasks to approximately 0° C, and set aside two of the flasks for blank determinations. Using a 50-ml graduate cooled by rinsing with portions of the cooled sample, pour 50 ml of the sample into each of the other two flasks. Swirl the contents of the flasks to effect solution. Allow the four flasks to stand, at room temperature for at least 10 minutes but not more than 30 minutes with occasional swirling. Add 10 ml of water previously cooled in an ice bath to each flask. Immediately titrate the contents of each of the flasks with 0.1*N* iodine solution until the yellow color begins to persist, add 5 ml of freshly prepared 1.0-percent aqueous starch indicator, and continue the titration to the first permanent blue end-point.

4.5.4.3 *Calculation.*

Percent acetaldehyde =

$$\frac{(B - A)\ N \times 2.2}{50 \times G}$$

where: A = ml 0.1*N* iodine solution required for sample.
B = average ml 0.1*N* iodine solution required for blank.
N = normality of iodine solution.
G = specific gravity of sample.

4.5.5 *Chlorides (inorganic).* Place 10 ml of water in a 250-ml glass-stoppered erlenmeyer flask and cool the water in an ice bath. Add approximately 10 ml of chilled ethylene oxide sample, 10 drops of concentrated nitric acid, and about 10 ml of approximately 0.1*N* silver nitrate solution. Report the presence of inorganic chlorides if a precipitate is formed.

4.5.6 *Acetylene.*

4.5.6.1 *Reagents.*

4.5.6.1.1 *Mixed indicator.* Dissolve 0.10 g of methyl red and 0.050 g of methylene blue in 100 ml of 95-percent ethyl alcohol conforming to grade I, class B, of Specification O-E-760. Mix the solution thoroughly and store it in a dark-colored bottle.

4.5.6.1.2 *Silver nitrate solution.* Dissolve 25 g of silver nitrate in 150 ml of water contained in a 1000-ml volumetric flask. Mix the solution thoroughly and dilute to the mark with 95-percent ethyl alcohol conforming to grade I, class B, of Specification O-E-760. Mix the solution thoroughly.

4.5.6.2 *Procedure.* Transfer 150.0 ml of the silver nitrate solution to each of two 250-ml glass-stoppered erlenmeyer flasks. Place the flasks in a suitable ice bath until the temperature of the contents is 2° ± 2° C. Add 7 to 9 drops of the mixed indicator to each of the flasks and, if necessary, neutralize the solution to a gray-green color with 0.02*N* sodium hydroxide, if acid, or with 0.02*N* nitric acid if alkaline. Remove one of the flasks from the ice bath for the blank. Using a chilled graduate, pour 50 ± 1 ml of the sample into the flask remaining in the ice bath. Swirl the flasks and compare the colors of the contents. Report the presence of acetylene if the solution containing the sample turns to a purple color. (If a positive test is obtained, destroy the silver acetylide by adding an excess of acid ferrous sulfate solution to the contents of the flask before discarding the solution.)

4.5.7 *Non-volatile residue.* Measure approximately 100 ml of the cold sample into a chilled 100-ml graduate and place it in a tared boro-silicate glass evaporating dish. Allow the ethylene oxide to evaporate to dryness at room temperature. Place the dish

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:09Z
Check the source to verify that this is the current version before use.

in an oven at 105° to 110° C. for 30 minutes. Cool the dish in a desiccator and reweigh it to the nearest milligram. Calculate the nonvolatile residue by subtracting the tare weight from the weight of the dish containing the residue.

4.5.8 *Color.*

4.5.8.1 *Standards.*

4.5.8.1.1 *Stock standard.* Dissolve 1.245 g of potassium chloroplatinate (K_2PtCl_6) and 1.000 g of cobaltous chloride ($CoCl_2.6H_20$) in water. Add 100 ml of concentrated hydrochloric acid, and dilute the solution to 1,000.0 ml with water. (This solution has a color of 500-on the APHA and ASTM platinum-cobalt scales.)

4.5.8.1.2 *Test standard.* Dilute 3.0 ml of the stock standard to 100.0 ml with water. (This solution represents a color of 15 on the platinum-cobalt scale.)

4.5.8.2 *Procedure.* Fill one of a pair of matched tall-form 100-ml comparison tubes to the mark with the test standard. Chill the second matched tube and fill it to the mark with the chilled sample. Immediately compare the colors of the test standard and the sample by viewing vertically down through the tubes using a white background. Report the sample as passing the requirement if the sample is no darker than the test standard.

4.6 **Acceptance/rejection criteria.** If either sample fails to meet the requirements of this specification, the lot represented shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 **Packing, level C.** Ethylene oxide in the quantities specified in the contract or order shall be packed in accordance with Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

5.2 **Marking.** In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129, including precautionary, regulation and statute marking.

6. NOTES

6.1 **Intended use.** Ethylene oxide mixed with specified quantities of carbon dioxide or chlorofluoro hydrocarbons to remove the explosion hazard is intended for use as a biological decontaminant.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Unit quantity desired.

(c) Supplemental evidence required.

6.3 **Batch.** A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing process intended to make the final product substantially uniform.

6.4 **Explosibility.** Liquid ethylene oxide is stable to detonating agents, but the vapor will undergo explosive decomposition. Pure ethylene oxide vapor will decompose partially; however, a slight dilution with air or a small increase in initial pressure provides an ideal condition for complete decomposition. Copper or other acetylide-forming metals such as silver, magnesium, and alloys of such metals should not be used to handle or store ethylene oxide because of the danger of the possible presence of acetylene. Acetylides detonate readily and will initiate explosive decomposition of ethylene oxide vapor. In the presence of certain catalysts, liquid ethylene oxide forms a poly-condensate. This reaction is highly exothermic. If the heat of the reaction is not conducted through the walls of a closed container at a rate capable of maintaining an equilibrium temperature, an increase in pressure results with an increase in reaction rate, leading to explosive conditions. Acid salts, such as stannic chloride and zinc chloride, and bases, such as alkali metal hydroxides, either solid or in aqueous solution, and tertiary amines are all effective catalysts. It is, therefore, imperative that the concentration of such contaminants be kept at a minimum when transporting or storing sizeable quantities of ethylene oxide.

6.5 **Toxicity.** Liquid ethylene oxide, con-

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

centrated or dilute, when exposed to the skin can cause severe delayed burns. Short exposures produce mild first degree burns, but prolonged exposures produce second degree burns with the formation of large blisters. Exposure to the vapor results in systemic manifestations and irritation to the respiratory system. Inhalation of ethylene oxide vapors, if prolonged, results in severe systemic poisoning with the symptoms of nausea, vomiting, headache, dysnea, and diarrhea. The anesthetic properties are similar to chloroform, but with pronounced undesirable side and after effects. Tests on guinea pigs have shown that a 5 percent by volume concentration of ethylene oxide vapor kills in a short time. The maximum concentration for 60 minutes without serious disturbance was 0.3 percent by volume.

6.6 Supersession data. This specification supersedes Chemical Corps Purchase Description 196-21-62, dated 15 January 1953.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any invention that may in any way be related thereto.

Custodians:
Army—CmlC
Navy—Docks

Preparing activity:
Army—CmlC

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:09Z
Check the source to verify that this is the current version before use.